

Lab 06 Instructions

Due March 10 5:00 PM PST

STAT240 Spring 2026

2026-03-06

For this homework, provide a single rendered Quarto Markdown file in pdf format on crowdmark for the problems (you may render the QMD file to html, and then convert the html file to pdf using the print function on your web browser). You can use the template `Lab_06.qmd`. **Provide in the markdown the final version of all of the code you wrote for this homework, and make sure long lines of code are wrapped in the rendered pdf. Lines which are cut off cannot be graded. If your Markdown file involves examining a large dataset, do not print the entire dataset to the markdown file in the steps of your solution (instead, suppress the output, or only show a small section of the data as an example).**

Problem 1

In this problem we will use data from the 1971 Canadian census, which looks at the relationship between the average education and income of people in 100 different professions. This data is included in the R package `carData` which you will need to install. Then, this data, in a data frame `Prestige`, can be loaded in using the following code:

```
library(carData)
data("Prestige")
```

Here we are only interested in the columns `education`, and `income`. We want to predict the value of `income` using `education` as the prediction.

Problem 1a

We will first split the data into a training and a test set, with ~75% in the training set and ~25% in the test set. Use `set.seed(100)`. Write some code to do this, calling each the `training_set` and the `test_set` in your R code.

(2 points)

Problem 1b

Now fit a linear regression to this training data and compute the RMSPE when this regression is used to predict on the test data. Be sure to print out this RMSPE.

(3 points)

Problem 1c

Now we will tune KNN for this data, before later predicting for the test data. For K taking a range of possible values from $K = 1$ to $K = 50$, use 5-fold CV to determine which value of K gives the best CV-RMSPE on this data. Plot this result and clearly state the choice of K .

(4 points)

```
library(caret) ## this package is needed
```

Loading required package: ggplot2

Loading required package: lattice

Problem 1d

For the best K in the previous part, use this choice of K to compute the RMSPE on the test data. Does this model give better predictions than linear regression?

(1 points)